

Hybridization of Porsche Panamera 970

TME095- Electric and hybrid vehicles

A project in fall of 2024 by Nihar Prakash Mehta, Ramkumar Huleppa Munavalli and Wilhelm Hansson.

Abstract

This assignment explores the development of a custom driving cycle, as well as focusing on hybridizing the vehicle powertrain and reducing fuel consumption. The key test includes developing the conventional powertrain and then hybridizing the vehicle to compare the fuel consumption from conventional to hybrid vehicle. To perform this project, MATLAB/Simulink and the QSS Toolbox were used.

Introduction

Our team (Group – 15) was assigned with Class -F, Luxury car segment. We built the conventional power train by taking required data from the Porsche Panamera and fuel consumption was noted down. The initial Porsche Panamera model was then hybridized in two different scenarios', series-hybrid and parallel-hybrid and the fuel consumption was calculated for these configurations. The fuel consumption after hybridizing was less compared to that of the conventional powertrain.

The report summarizes the methodology and highlights the key observation of reduced fuel consumption in hybrid vehicles compared to an ICE vehicle.

Vehicle Parameters & Assumptions

The vehicle parameters considered for Porsche Panamera are listed below in the table. For reference we have considered the data of the Porsche Panamera 970 from *Carinf*.

Parameters	Conventional	Series-Hybrid	Parallel-Hybrid
Engine capacity (KW)	285	300	285
Gearbox	Automatic	Simple transmission	Automatic
Fuel Type	Gasoline	Gasoline	Gasoline
Electric Motor (KW)	-	300	16
Electric Generator (KW)	-	300	-
Mass (Kg)	2081	2181	2181
Drag Co-efficient	0.3	0.3	0.3
Wheel diameter(m)	0.62	0.62	0.62
Vehicle cross section(m ²)	2.330	2.330	2.330

Table 1

Gear	Ratio
1 st gear	5.97
2 nd gear	3.22
3 rd gear	2.08
4 th gear	1.42
5 th gear	1.05
6 th gear	0.84
7 th gear	0.68
8 th gear	0.53
Differential gear	3.36
Simple transmission	2

Table 2

Method

Considering our vehicle class, we were looking at not only testing the performance capabilities of our vehicle but also implementing it into a driving scenario that shows how the vehicle is normally driven. We first tested the conventional model in the acceleration performance mode of QSS. The conventional Porsche Panamera operates in the most efficient region of the engine at slightly higher speed/torque values. However, it is not always possible to drive the vehicle this way, and in practical scenarios the vehicle operates more around lower speeds. Our driving cycle highlights such a scenario to analyze the fuel consumption by implementing and comparing a series and parallel hybrid version of this vehicle with the provided battery requirements. The series and parallel hybrid for this vehicle were modelled on QSS and the fuel consumption was checked by ensuring that at the end of the cycle the vehicle SOC is the same as the initial SOC. This gives us a fair comparison on fuel consumption for the three models.

Goals:

- Performance test: Get similar results to the acceleration performance of the original vehicle. 0-100 km/h in 6.8 seconds and 0-160 km/h in 14 seconds.
- Better fuel consumption for the drive cycle using hybrid models
- Battery controller should be able to pick and operate the engine/generator/electric motor and make it operate at the best efficiency and best fuel consumption for a given power demand.

Conventional Setup

The conventional setup is modelled as shown in the figure. We use an Automatic gear box. The engine follows the otto-cycle powered by Gasoline.

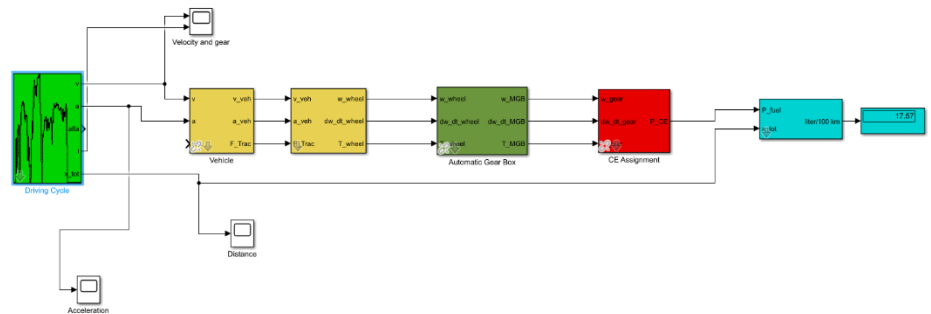


Figure 1

Series Setup

For the series configuration, the EM can be powered by the ICE connected to the generator as well as the battery. It can also be powered by a combination of the two.

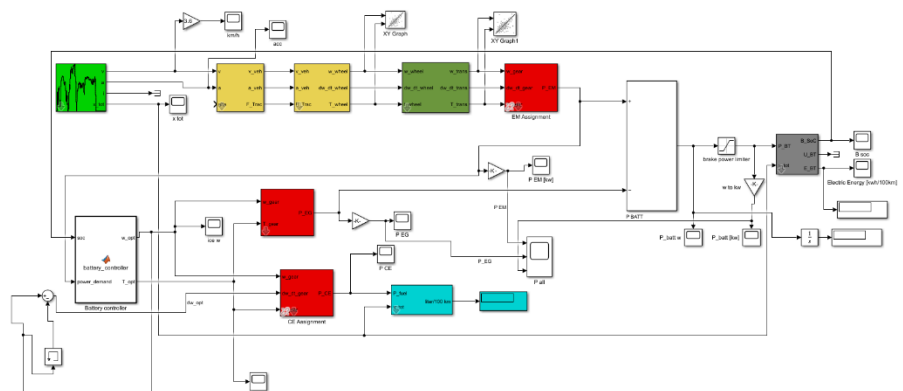


Figure 2

Parallel Setup

In the parallel setup, we have the IC Engine (285 kW) and the Electric Motor (16 kW) giving a combined power of around 300 kW. The same battery which is used in series hybrid is used here. The Automatic gearbox feeds to signals to the battery controller. Based on the requirement and conditions we define, the battery controller feeds the signal to the required energy source/s.

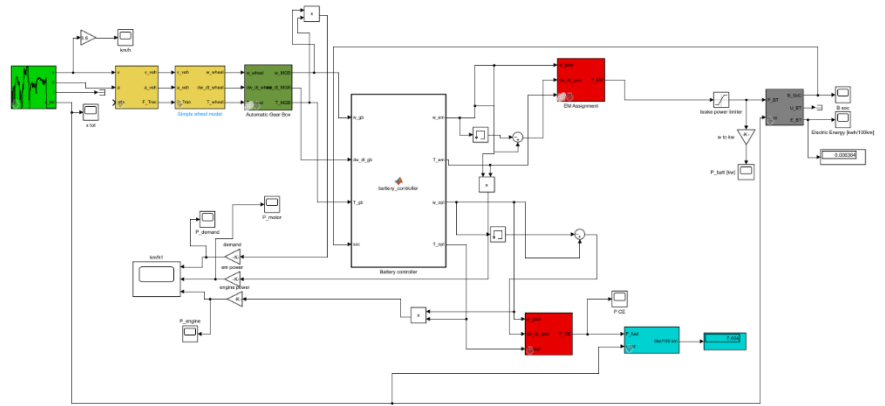


Figure 3

Design layout

The Porsche Panamera 970 is a class F luxury car with high performance capabilities. It can accelerate from 0 - 100 km/h in about 6.8 seconds. The properties of the Porsche Panamera can be found in *table 1* and *table 2*.

Our defined driving cycle highlights the general usage of a vehicle of this class. This includes some urban driving, as well as highway driving.

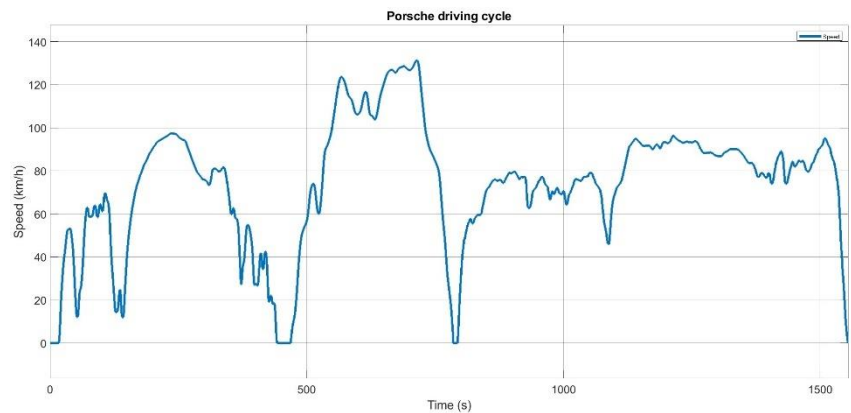


Figure 4

With background to the original configuration of the Porsche Panamera, the hybridization of it requires a battery. The following specifications seen in *table 3* are used for the battery.

Battery weight

Battery max capacity[kWh]	Density Li- ion [Wh / kg]	Cell capacity [Ah]	Cell voltage [V]	Max peak power output [A]
2.5	200	15	3.8	+ - 100

Table 3

It's also necessary to define the number of cells in series and parallel with respect to the max capacity of the battery, 2.5 kWh.

$$\text{Energy per cell} = 15 * 3.8 = 57 \text{ Wh/cell}$$

$$\frac{\text{Max energy}}{\text{energy per cell}} = \frac{2500}{57} = 43.8 \rightarrow 43 \text{ cells}$$

This results in 43 cells in series and 1 cell in parallel.

Density for a Li- ion cell is in the range of 200 – 300 Wh / kg. 200 Wh / kg was used for our case.

$$\frac{\text{Energy}}{\text{Energy density}} = \frac{2500}{200} = 12.5 \text{ kg}$$

This mass doesn't account for cooling, electric components our housing. Therefore, the total mass of the battery is including these factors, and is set to 100 kg

Energy management for series

For the series hybrid, the electric motor can draw power from either the battery, an ICE connected to a generator or from both sources at the same time. The power delivered by the battery is limited by how the battery is configured. 43 cells in series and 1 in parallel result in the battery having a power of

$$P_{batt} = 3.8 \text{ V} * 43 \text{ cells} * 100 \text{ A} = 16.3 \text{ kW}.$$

The SOC of the battery is managed by a battery controller. The controller has three main purposes:

- Respect SOC limits, lower SOC limit = 20%, upper SOC limit = 90 %.
- Utilize the battery efficient and as much as possible to minimize fuel consumption
- Control the limit of how much power the battery can charge / discharge with.

For the controller to be useful not only to control the battery, but also with respect to fuel consumption, it needs to control how and when the ICE is used. It is completely turned off below a certain power level, 9 kW, which the battery can sustain the EM with by itself. When the power demand exceeds this power level of 9 kW, the ICE is turned on to meet power demand. Charging the battery happens either when the ICE can operate in its most efficient region, seen in *figure 8*, or when the SOC drops to 20%. *Figure 8* also shows where the highest efficiency of the EGU is found, around ~ 120 kW. *Figure 9* shows how the BSFC (Break specific fuel consumption) gets lower as the ICE's efficient regions are approached.

To achieve an efficient use of the battery, the controller uses an on- off strategy for the SOC. Depletion until 20%, then recharge to 40%. This is seen in *figure 5*.

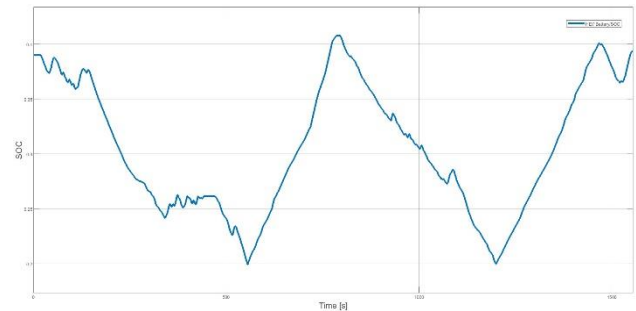


Figure 5

Energy management for parallel

The controller for the parallel is configured in a way that utilizes the benefit of the vehicle being powered by the Electric Motor, the IC Engine or both together. It is designed to help operate the IC Engine and the Electric motor in the most efficient region possible for the drive cycle. The power limitations on the battery remain the same as we use the same battery in the parallel hybrid.

The energy management is done using the battery controller which controls the mode of operation of the powertrain based on the power demand and SOC. It operates mainly on these conditions:

- The SOC must always be within 20% and 90%. If the SOC reaches 90% we use battery power to drive the Electric Motor, and if it approaches 20%, we charge the battery and meet the power demand, using the IC engine.
- The mode is also decided based on power demand. For power demands within the battery power limitations, the vehicle is powered completely by the battery and the ICE is switched off.
- When we have power demands higher than the battery limitation but lower than efficient operating regions of the engine, we keep the motor on and meet the remaining power demand with the IC Engine.
- For higher power demands where the IC Engine is the most efficient, we turn off the Electric Motor and only use the IC Engine.
- For the highest power demands, we operate with both the IC Engine (max power 285 kW) and Electric Motor (max power 16 kW) giving us a total power of roughly 300 kW. Like the series, we restrict the power from the electric motor to 9kW here as well.
- The controller also limits the amount of charge/discharge from the battery based on power limitations on the battery.
- Here as well, like the series, we use the on-off strategy to effectively use the battery

Test and Results

The testing is done on the same driving cycle so that we can compare the results.

Conventional Test Results

Our vehicle's conventional model was able to do 0-100 km/h in 6 seconds and 0-160 km/h in 16 seconds on the performance test. When we ran it for the driving cycle, the engine map looks as shown below. We get a lot of points operating in poor fuel consumption areas due to the low power demand of the driving cycle.

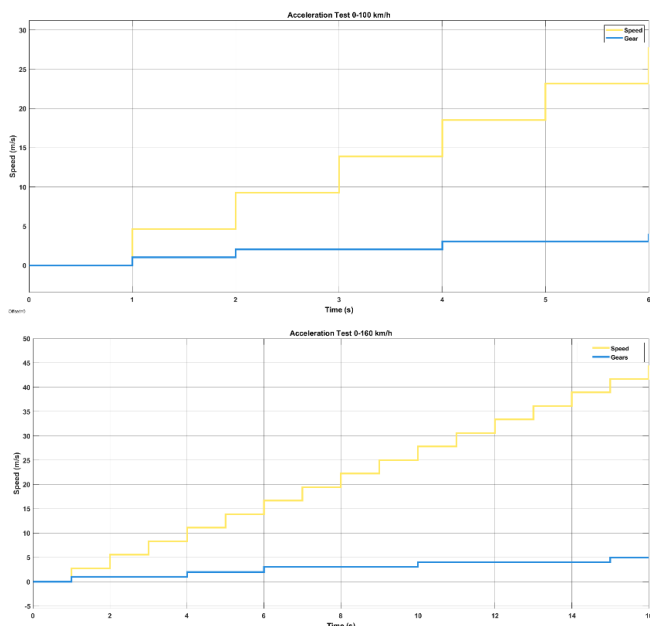


Figure 6

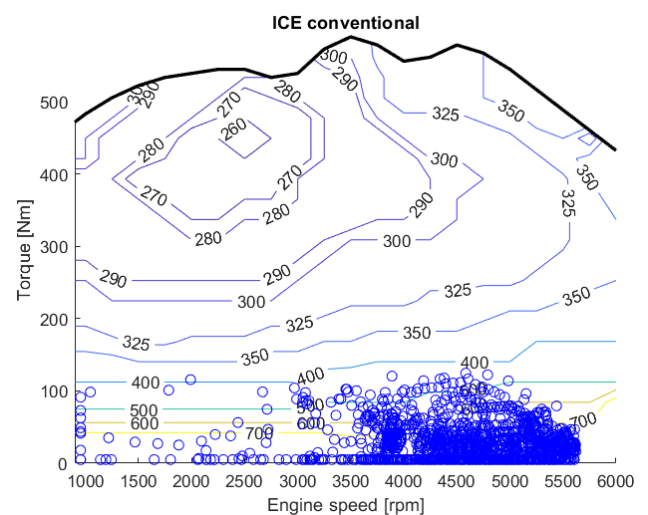


Figure 7

Series Test Results

For the series configuration, *figure 11* shows how the power, respectively from the battery and the ICE/ generator combination, is calculated based on the power demand required from the EM. The battery (orange curve) provides power alone for lower power demand, as well as supporting the ICE/generator for higher power-demand. The total fuel consumption of the driving-cycle is 11.29 liter/100 km. Using the battery to a high extent and charging it when the ICE can operate in efficient regions, seen in *figure 9*, seems to decrease the total fuel consumption during the driving cycle. When power demand is negative, the battery is charged.

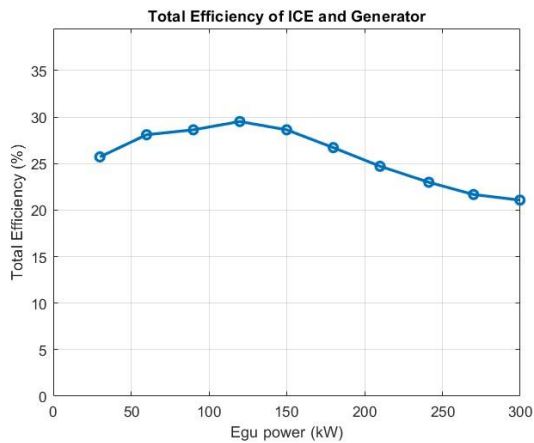


Figure 8

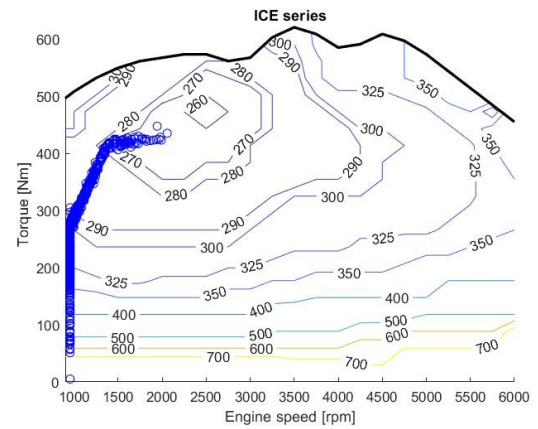


Figure 9

The operating points of the EM are seen in *figure 10*.

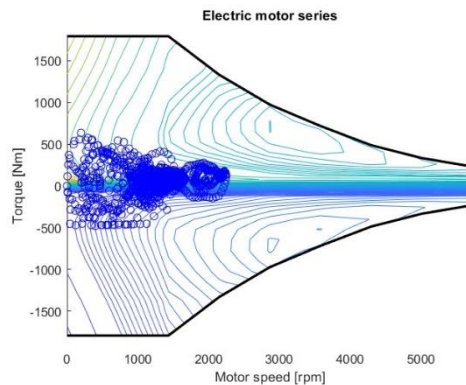


Figure 10

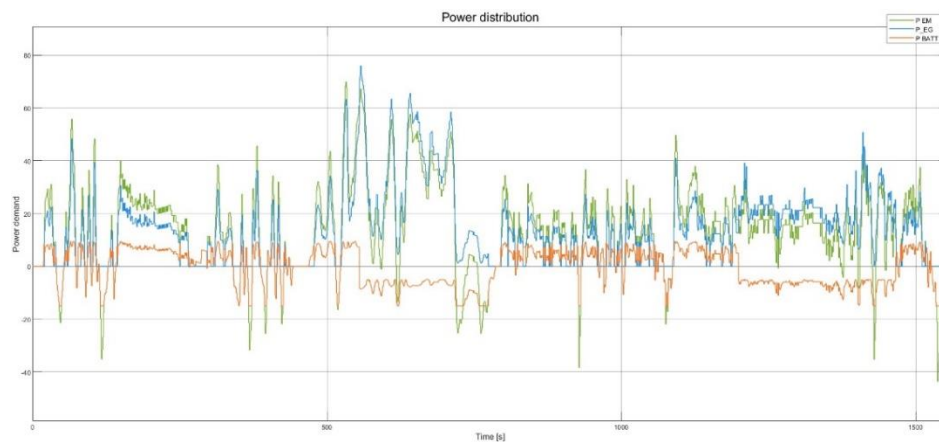


Figure 11

Parallel Test Results

The ICE and EM maps are shown below for the parallel hybrid. The car operate in the most efficient regions of the ICE and EM for the given driving cycle. We don't reach more efficient/high power demand regions as our driving cycle does not demand it from the engine. The initial SOC was 28% and the final SOC was 28.9%. The fuel consumption for parallel is 7.404 liter/100 km.

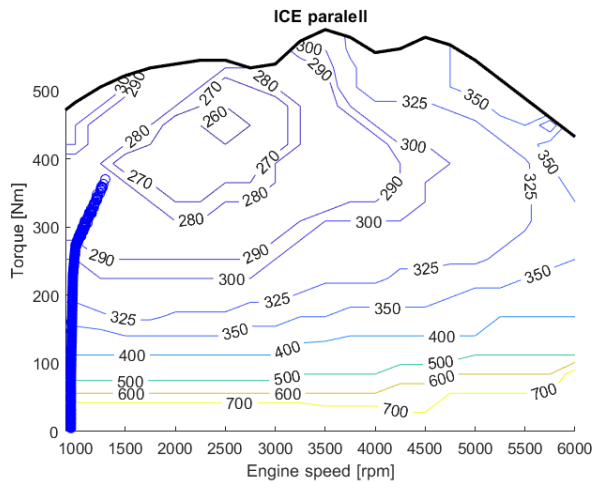


Figure 12

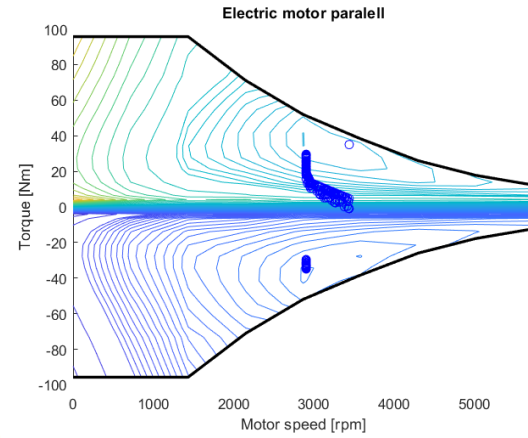


Figure 13

In figure 14, we see the power distribution for this model. At any given instant, the power demand is met by either the ICE, the Electric motor or both. This is also seen from the instances when power demand is negative (regeneration). The power demand at any instant is basically power from the ICE + power from the electric motor.



Figure 14

Comparison of models

Table 4 shows that fuel consumption decreases as the conventional vehicle gets hybridized. The parallel lowers the fuel consumption by **57.85%** and series by **35.74%**. This is mainly due to higher mechanical losses in the series configuration as there are more steps of energy conversion compared to the parallel where it directly propels the wheels.

Model	Conventional	Series	Parallel
Fuel Consumption (liters/100km)	17.57	11.29	7.404
Improved fuel consumption (%)	-	35.74	57.85

Table 4

Conclusions

Let us look at the Comparison of models (Table 4) to draw a better conclusion. It is to be noted that we have considered only the custom drive cycle for comparison in Table 4.

We can see from the series test results (figure 10) that the battery can be used to a high extent and charging it when the ICE is running in the efficient regions due to this the fuel consumption is lowered by **35.74%** when we compare it with the conventional powertrain. The parallel test results (figure 11,12 &13) demonstrate that the ICE and EM operate in the most efficient regions for the custom driving cycle and the battery is used as much as possible through the on- off strategy through the controller. Due to this fuel consumption is lowered by **57.85%** compared to the conventional powertrain.

As a team we learned the importance of making decisions in building the conventional powertrain with the custom driving cycle and available driving cycles in the QSS tool- box. Optimizing the fuel consumption by hybridizing the car from conventional to series-hybrid and parallel-hybrid by modeling and simulating various powertrain configuration has a great impact on the fuel consumption. The challenging part was to design the controller unit to hybridize the car and we explored many different ways of implementing the controller logic to make it function as we wanted it to.

There are also a few weaknesses in the model that would require further effort in the future:

- Interpolation errors
- Battery power output is limited to 9 kW even though it can produce 16 kW. This was done due to sometimes unexpected drop in voltage.
- Battery weight might be a bit on the heavier side. Hard to estimate the weight for the housing, cooling unit etc.
- A gearbox controller for the parallel model might give a more optimized result.

References

Carinf.com. Downloaded 15/20-2024 from <https://www.carinf.com/en/f270418789.html>.